

MHDRM Cross-Site Readiness Assessment

Prototype screening report for multimodal federated learning

Assessment Context

Project name	Scenario 2: AI-enabled drug discovery
MMFL use case	Drug discovery of a potential new antipsychotic medication.
Target outcome or label	Continuous measurement of whether the candidate compound demonstrates activity against the selected therapeutic target.
Participating sites	Pharmaceutical company 1 Pharmaceutical company 2 University research department 1 University research department 2 University research department 3
Target population	Unspecified, generally applies to patients in need of antipsychotic medication.
Required modalities	Genomic data, Structured EHR data, Other

Assessment Summary

Total score	Readiness	Classification
1/14	7.1%	Substantial preparation needed

Pillar Results

Category	MHDRM pillar	Response	Score	Flag
Data Characteristics	Modality Availability	N/E — Insufficient evidence	N/E	Evidence gap
Data Characteristics	Data Quality and Completeness	N/E — Insufficient evidence	N/E	Evidence gap
Data Characteristics	Population Representativeness	0 — Not demonstrated	0	Critical gap
Integration Readiness	Standardization and Interoperability	0 — Not demonstrated	0	Critical gap
Integration Readiness	Cross-Modal Alignment	0 — Not demonstrated	0	Critical gap
Organizational Readiness	Privacy and Security by Modality	1 — Partially demonstrated	1	
Organizational Readiness	Provenance and Governance	0 — Not demonstrated	0	Critical gap

Evidence and Rationale

MHDRM pillar	Evidence reviewed and scoring rationale
Modality Availability	Modality coverage and sample sizes have not yet been evaluated across sites.
Data Quality and Completeness	Data quality expectations have not been established. They have not yet completed full data assessments.
Population Representativeness	Target populations and relevant clinical subgroups are not defined.

MHDRM pillar	Evidence reviewed and scoring rationale
Standardization and Interoperability	Sites use different chemical identifiers, molecular representations, and phenotype definitions which they have not shared yet.
Cross-Modal Alignment	Genomic and structured clinical data are available at the university sites, while molecular graph data are available at the pharmaceutical companies. No common method has been established for linking genomic and clinical phenotypes or relating those phenotypes to the molecular representations used in the proposed learning objective.
Privacy and Security by Modality	Each organization has internal security mechanisms in place, but they have not assessed the risks associated with combined modalities.
Provenance and Governance	Data use agreements are still in development, and governance responsibilities have not been assigned.

Potential critical gaps: Population Representativeness, Standardization and Interoperability, Cross-Modal Alignment, Provenance and Governance

Evidence gaps: Modality Availability, Data Quality and Completeness

Prototype limitation: This report provides a preliminary preimplementation screening of cross-site data compatibility. The scores and thresholds have not been empirically validated and do not certify that an MMFL initiative will be successful.